Homework3

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```r
library(tidyverse)
library(here)
library(janitor)
library(readxl)

kelp <- read.csv(here("data","temp-kelp.csv"))

my_data <- read_excel(here("data", "my_data.xlsx")) |> clean_names()
```

Problem 1

a. The two appropriate tests would be a correlation test and a linear regression. A correlation test measures the strength and direction of the relationship between ocean temperature (°C) and giant kelp frond elongation rate (cm day ^{1}), while a linear regression examines how changes in ocean temperature predict changes in kelp growth rate. Correlation only measures association between the two variables, but regression models the relationship and identifies temperature as the explanatory variable and elongation rate as the response variable.

b.

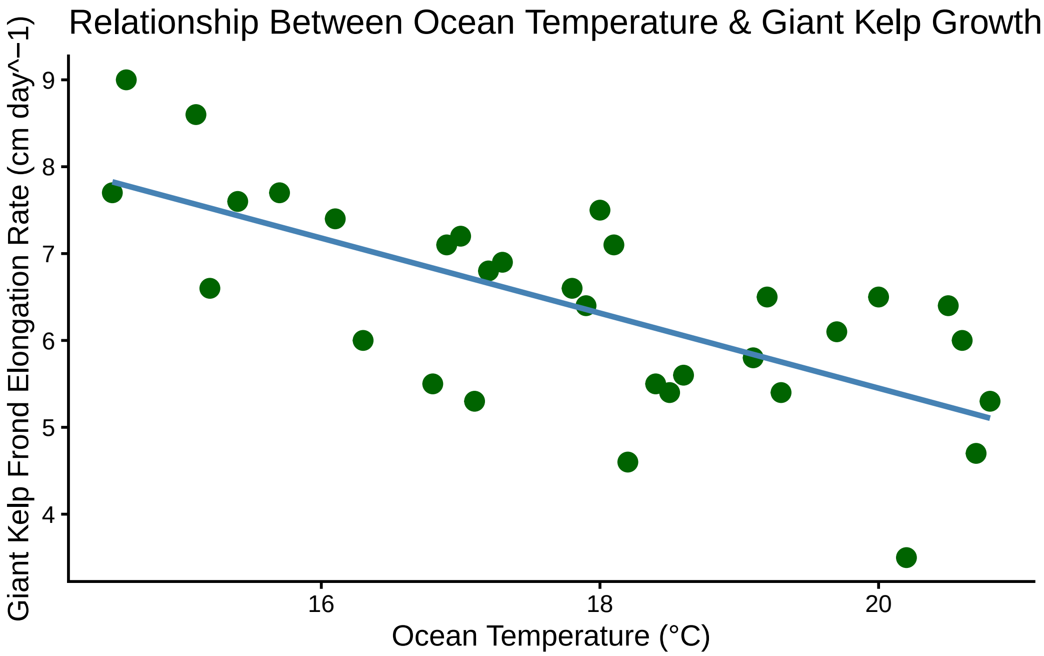
```r
ggplot(kelp, aes(
            x = temp_c,
            y = kelp_elong)) +

  geom_point(color = "darkgreen", size = 3) +

  geom_smooth(method = "lm", color = "steelblue", se = FALSE) +

  labs( x = "Ocean Temperature (°C)",
        y = "Giant Kelp Frond Elongation Rate (cm day^-1)",
```

```
      title = "Relationship Between Ocean Temperature & Giant Kelp Growth") +
  theme_classic()
```

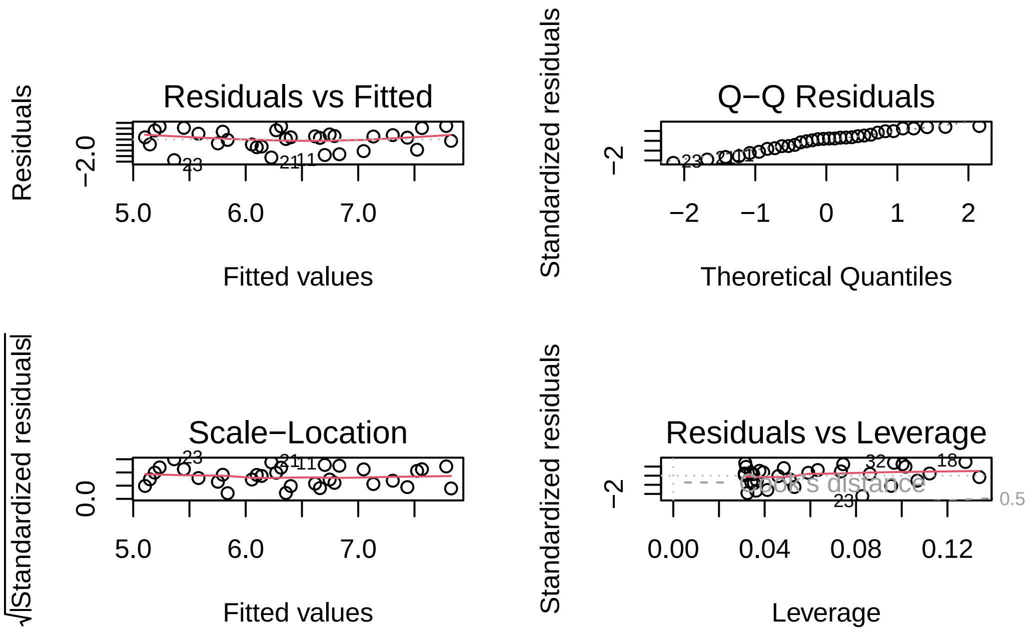


c. To evaluate the relationship between ocean temperature and giant kelp frond elongation rate, I checked the assumptions for a linear regression. Specifically, I checked for linearity, normality of residuals, and equal variance of residuals. I assessed these assumptions using a scatterplot with a regression line and diagnostic plots from the regression model. Based on the plots, the assumptions appeared reasonably met because the relationship looked linear and the residuals did not show major deviations from normality or equal variance.

```
# Create diagnostic plots

kelp_lm <- lm(kelp_elong ~ temp_c,
              data = kelp)

par(mfrow = c(2, 2))
plot(kelp_lm)
```




```r
# Display the regression results
summary(kelp_lm)
```

```

Call:
lm(formula = kelp_elong ~ temp_c, data = kelp)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8643 -0.5017  0.1790  0.5659  1.2177 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473, Adjusted R-squared:  0.4554 
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

d. To evaluate the strength of the relationship between ocean temperature and giant kelp frond elongation rate, I used a linear regression because both variables are continuous and I was interested in determining whether temperature could predict kelp growth rate.

The linear regression showed a relationship between ocean temperature and giant kelp frond elongation rate (report the slope coefficient, (R^2), and p-value from your output here). This suggests that as ocean temperature changes, kelp elongation rate also changes in a predictable way.

e. The results of this test suggest that ocean temperature is related to giant kelp frond elongation rate, meaning that changes in temperature may influence how quickly giant kelp grows. If warmer temperatures are associated with reduced growth rates, this could indicate that continued ocean warming from climate change may negatively impact giant kelp forest health and productivity. These findings highlight the importance of monitoring ocean temperature trends and protecting kelp ecosystems that may be vulnerable to warming conditions.

f.

```r
# Run a correlation test

cor.test(kelp$temp_c, kelp$kelp_elong)
```

```

    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

The correlation test and the linear regression would likely lead to the same decision about the null hypothesis because both tests evaluate the relationship between ocean temperature and giant kelp frond elongation rate. However, the correlation test measures the strength and direction of the association between the two continuous variables using the correlation coefficient (r), while the linear regression additionally models how temperature predicts changes in elongation rate through the regression slope. As a result, both tests should provide similar interpretations about whether the relationship is weak or strong, positive or negative, and statistically significant.